Mechanisms of acute ankle syndesmosis ligament injuries in professional male rugby union players: a systematic visual video analysis

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ABSTRACT

Background The mechanisms of acute ankle syndesmosis ligament injuries in professional rugby union are not well understood.

Aim To describe the mechanisms of acute ankle syndesmosis ligament injuries in male professional rugby union players using systematic visual video analysis.

Methods All time-loss acute ankle syndesmosis ligament injuries identified via retrospective analysis of the Leinster Rugby injury surveillance database across the 2013/2014 to 2017/2018 seasons were considered as potentially eligible for inclusion. Three Chartered Physiotherapists (analysts) independently assessed all videos with a final consensus meeting convened to describe the injury mechanisms. One of the analysts developed a qualitative description of each injury mechanism.

Results Thirteen acute ankle syndesmosis injuries were included in this study. The most common biomechanical presentation of the injured limb was characterised by hip flexion, knee flexion, knee valgus, ankle dorsiflexion, pronation and external rotation of the foot. Six of the included injuries (46%) were the result of an external load application to the injured limb (ie, direct contact injuries). In the other seven instances (54%), the injury was the result of a collapse of the injured player's body mass over the injured ankle joint, while tackling or being tackled, without direct contact to the injured limb (ie, indirect contact injuries).

Conclusion Injuries incurred while tackling were exclusively the result of suboptimal tackle mechanics. A majority of injuries incurred while being tackled involved a posterior tackle, which often resulted in a posterior collapse of the injured player's centre of mass over a fixed externally rotated foot.

INTRODUCTION

The World Rugby (formerly International Rugby Board) World Cup injury surveillance studies illustrate that ankle joint injuries are among the most commonly sustained lower limb musculoskeletal injuries during international tournaments.[1–3] Although the largest percentage of ankle joint injuries sustained in professional rugby union are lateral ligament sprains, ankle syndesmosis ligament injuries are not uncommon.[4]

Understanding the sport-specific mechanisms of injury associated with prevalent musculoskeletal injuries in a particular sport is fundamental to the success of injury prevention initiatives.[5] Technologies such as video analysis[6 7] and lower limb kinematic analysis,[8–11] as well as novel biomechanical techniques, including model-based image matching,[12 13] can facilitate an enhanced understanding of sport-specific mechanisms of injury. A number of studies have detailed the mechanisms of injury associated with lateral ligament sprains of the ankle joint.[6–10 12 13] The observations of these studies illustrate that the mechanisms of injury include sudden high-velocity, high-magnitude inversion and internal rotation of the ankle/foot. The mechanisms of injury associated with ankle syndesmosis ligament injuries are less clear, but have been reported to include external rotation of the foot, eversion of the talus within the ankle mortise and excessive dorsiflexion.[14]

To date, to our knowledge, no studies have been published that detail the mechanisms of ankle syndesmosis ligament injuries in male professional rugby union. Therefore, the objective of our study was to describe the mechanisms of acute ankle syndesmosis ligament injuries in male professional rugby union players using video analysis.

METHODS

Injury inclusion

All time-loss acute ankle syndesmosis ligament injuries identified via retrospective analysis of the Leinster Rugby injury surveillance database across the 2013/2014 to 2017/2018 seasons and confirmed by MRI were considered as potentially eligible for inclusion. Leinster Rugby is a professional rugby union team registered in the Ireland, who participate in the PRO14 and European Rugby Champions Cup. In total, 30 time-loss ankle syndesmosis ligament injuries were registered on the injury surveillance database across these five seasons. We accessed the Leinster Rugby match and training video database to retrieve video footage of the match or training session during which each injury was incurred. One of the authors (AB) used data from the injury surveillance database (time of injury during match or training) to identify each injury event. Video footage was available for 27 of 30 time-loss ankle syndesmosis injuries. Initial video review revealed that the mechanism of injury was not visible in 14 instances, resulting in the inclusion of 13 acute ankle syndesmosis ligament injuries for quantitative and qualitative analysis. Additional details on MRI findings, treatment type (conservative vs surgery) and time-to-return to unrestricted training were also documented for each injury.

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Ethical approval was received from the University College Dublin Human Research Ethics Committee (LS-19-18-Delahunt).

Video processing

Video footage of the specific matches and training sessions during which the injuries were incurred were converted to and saved as MPEG-4 files, which allowed frame-by-frame review using QuickTime 7 (Apple, Cupertino, California, USA). For cases where two or more camera views were available, we made one single video with all sequences synchronised side-by-side.

Video analysis

Three Chartered Physiotherapists (analysts) participated in the video analysis (AB, GF, ED). These authors developed an injury assessment form that included specific domains common to previously published injury mechanism video analysis studies.[7 15 16] This injury assessment form was pilot tested by one of the other authors (FK) prior to formal video analysis and data extraction. The form enabled us to characterise the injury situation and (injured) player actions at the time of injury (tables 1 and 2). Additionally, an assessment of ipsilateral lower limb and trunk biomechanics at the index frame (the probable injury event) was also detailed for each injury (table 3). The analysts independently assessed all videos in real time and then frame-by-frame to record the estimated time point of initial contact (as all injuries were sustained during contact), as well as the index frame. As per a previous video analysis study, we defined a contact injury as one whereby the injured limb was subjected to a directly applied external load.[15] We defined an indirect contact injury as one whereby there was contact with any other body region other than the injured leg.[15] Following the completion of this initial step, a meeting was convened during which the analysts discussed and reached a consensus on the details of the injury assessment forms and the whole body biomechanics of the injured player at the index frame. The results of the consensus reached by the three Chartered Physiotherapists were then reviewed and discussed with the Leinster Rugby contact skills coach. Additionally, one of the analysts (ED) developed a qualitative description of each injury mechanism that was approved by all other authors (online supplemental file 1). Images of a number of the injuries are included as a supplementary file (online supplemental file 2).

Table 1 Injury situation details and (injured) player actions at the time of injury

Injuries sustained while tackling					
Case	**Player position**	**Injured limb**	**Location on field**	**Playing situation**	**Player activity**
#1	Back row	Left	Between own 22 m line and 10 m line	Defensive	Injury was likely sustained while tackling an opposition player to the ground
#5	Back row	Left	Just outside the opposition team's 22 m line	Defensive	Injury was likely sustained while tackling an opposition player to the ground
#7	Front row	Right	During a small-sided training match	Defensive	Injury was likely sustained when a player from his own team fell onto his leg/foot
#10	Centre	Right	Between the opposition team's 10 m line and 22 m line	Defensive	Injury was likely sustained when a player from own team fell onto his leg/foot during a twoplayer tackle on an opposition player
#11	Centre	Right	Between half-way line and own 10 m line	Defensive	Injury was likely sustained while tackling an opposition player, whose momentum resulted in a backward collapse of the (injured) player
#13	Centre	Right	Just outside the opposition team's 22 m line	Defensive	Injury was likely sustained while tackling an opposition player to the ground

Table 2 Injury situation details and (injured) player actions at the time of injury

Injuries sustained while being tackled					
Case	**Player position**	**Injured limb**	**Location on field**	**Playing situation**	**Player activity**
#2	Winger	Left	At own 22 m line	Defensive	Injury was likely sustained while being tackled to the ground by an opposing player
#3	Back row	Right	At the opposition team's 5 m line	Offensive	Injury was likely sustained while being tackled to the ground by an opposing player
#4	Second row	Right	At own 10 m line	Offensive	Injury was likely sustained while being tackled to the ground by an opposing player
#6	Back row	Left	Outside own 22 m line	Offensive	Injury was likely sustained while being tackled to the ground by an opposing player
#8	Winger	Left	Outside the opposition team's 22 m line	Offensive	Injury was likely sustained while being tackled
#9	Back row	Right	Between the opposition team's 10 m line and 22 m line	Offensive	Injury was likely sustained while being tackled to the ground by an opposing player
#12	Out half	Right	During a small-sided training match	Offensive	Injury was likely sustained while being tackled to the ground by an opposing player

RESULTS

Thirteen acute ankle syndesmosis injuries were included in this study. Seven of the injuries were sustained by forwards (front row × 1; second row ×1; back row ×5), with the other six injuries sustained by backs (out half ×1; centre ×3; winger ×2). Of the 13 injuries, 8 involved the right ankle and 7 involved the left ankle. The anterior inferior tibiofibular ligament was injured in all 13 cases (100%). The posterior inferior tibifibular ligament was injured in nine cases (69%). The deltoid ligament was injured in four cases (31%). The anterior talofibular ligament and calcaneofibular ligament were injured in two cases (15%). Ten of the cases were managed surgically, with the other three cases managed conservatively. The mean time-to-return to unrestricted training was 61 days (SD=25 days).

Six injuries (46%) were incurred while tackling an opposition player with the remaining seven injuries (54%) incurred while being tackled by an opposition player. The most common

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Table 3 Ipsilateral lower limb and trunk biomechanics at index frame (the probable injury event)

Case	Hip sagittal plane	Knee sagittal plane	Knee frontal plane	Ankle sagittal plane	Foot frontal plane	Foot transverse plane	Trunk sagittal plane	Trunk rotation
#1	Unknown	Unknown	Unknown	Dorsiflexion	Pronation	External rotation	Unknown	Unknown
#2	Flexion	Flexion	Unknown	Neutral	Unknown	External rotation	Contralateral	Flexion
#3	Flexion	Flexion	Valgus	Dorsiflexion	Pronation	External rotation	Contralateral	Flexion
#4	Flexion	Flexion	Valgus	Unknown	Pronation	External rotation	Neutral	Flexion
#5	Flexion	Flexion	Valgus	Unknown	Pronation	External rotation	Neutral	Flexion
#6	Flexion	Flexion	Valgus	Dorsiflexion	Pronation	External rotation	Contralateral	Flexion
#7	Flexion	Flexion	Valgus	Dorsiflexion	Pronation	External rotation	Contralateral	Flexion
#8	Flexion	Flexion	Valgus	Dorsiflexion	Pronation	External rotation	Ipsilateral	Flexion
#9	Flexion	Flexion	Valgus	Dorsiflexion	Unknown	External rotation	Ipsilateral	Flexion
#10	Flexion	Flexion	Valgus	Neutral	Pronation	External rotation	Ipsilateral	Flexion
#11	Flexion	Flexion	Unknown	Dorsiflexion	Pronation	External rotation	Contralateral	Flexion
#12	Flexion	Flexion	Unknown	Dorsiflexion	Unknown	External rotation	Ipsilateral	Flexion
#13	Flexion	Flexion	Unknown	Dorsiflexion	Pronation	External rotation	Unknown	Flexion

biomechanical presentation of the injured limb at the index frame (the probable injury event) was characterised by; hip flexion, knee flexion, knee valgus, ankle dorsiflexion, pronation and external rotation of the foot. Six of the included injuries (46%) were the result of an external load application to the injured limb; typically via a tackle by an opposition player or due to inadvertent contact by a team mate, and were classified as direct contact injuries. In the other seven instances (54%), the injury was the result of a collapse of the injured player's body mass over the injured ankle joint, while tackling or being tackled, without direct contact to the injured limb (ie, indirect contact injuries).

Injuries incurred while tackling

Six of the 13 injuries (46%) were incurred while tackling an opposition player (cases #1, #5, #7, #10, #11 and #13). In all instances, the tackle mechanics of the (injured) player were suboptimal. In two of these instances (cases #5 and #11), the (injured) player 'dropped' to a one-knee position while making a front-on tackle. At the time of initial contact during both these tackles, the (injured) player's trunk was flexed at the hip joints, with concomitant knee flexion and ankle dorsiflexion. In case #5 (indirect contact injury), the forward momentum of the opposition player resulted in a posterior and lateral collapse of the (injured) player over his kneeling limb (left), with the injured ankle (left) being fixed in external rotation during the (probable) injury event. In case #11 (indirect contact injury), the forward momentum of the opposition player resulted in a posterior and lateral collapse of the (injured) player over his non-kneeling limb (right), with the injured ankle (right) joint being fixed in an externally rotated position during the (probable) injury event. In both case #1 (direct contact injury) and case #13 (indirect contact injury), the injury was also the result of a posterior collapse of the (injured) player's centre of mass over a fixed externally rotated ankle joint, with this injury mechanism being compounded in case #1 by the opposition player falling on the (injured) player's ankle (left) joint, thus, 'pinning' it in external rotation. In case #7 (direct contact injury), which was sustained during a small-sided training drill, the (injured) player's entry as a supporting player on the edge of a maul was suboptimal. He entered from a posterior–lateral position, which resulted in the lateral aspect of his right lower limb being oriented directly posterior to the ball carrier. When the ball carrier was tackled to the ground, the (injured) player's right knee joint was subjected to an externally induced valgus collapse, with the ball carrier 'pinning' his right foot in a pronated and externally rotated position. In case #10 (indirect contact injury), the (injured) player 'jumped' into the tackle from a lateral direction to halt the forward momentum of the attacking opposition player who was being concurrently tackled from behind by a team-mate of the (injured) player. While landing on his right foot, the (injured) player's right knee joint was subjected to an externally induced valgus collapse, as a result of impact by his team-mate. During this impact, his right foot was 'pinned' in external rotation as his team-mate fell onto his foot. In all tackling injury cases, pronation and external rotation of the foot relative to the tibia were characteristic features of the injury mechanism.

Injuries incurred while being tackled

Seven of the 13 injuries (54%) were incurred while being tackled by an opposition player (cases #2, #3, #4, #6, #8, #9 and #12.). In four of the seven cases (cases #2, #3, #9 and #12), the injury was the result of a posterior collapse of the (injured)

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player's centre of mass over a fixed externally rotated ankle joint. In case #2 (indirect contact injury), the (injured) player was tackled from a posterior-lateral direction around his waist on his right side by an opposition player. The opposition player's momentum resulted in the (injured) player being pulled to the ground. The (probable) injury event occurred during this tackle, while the (injured) player's torso was rotated away from the injured side (left ankle) and his centre of mass was collapsed posteriorly over a flexed knee joint and externally rotated (left) foot. In case #3 (indirect contact injury), the (injured) player was the front support lifter in a lineout. At the end of the lift he was tackled from behind by an opposition player, who made contact around his waist. His right foot was fixed in an externally rotated position due to interlocking with the opposition player's right lower limb. The (probable) injury event occurred during this tackle, while the (injured) player was being pulled to the ground; his torso was rotated away from the injured side (right ankle) and his centre of mass was collapsed posteriorly over a fixed externally rotated right foot. In case #9 (indirect contact injury), the (injured) player was accelerating in an offensive carry. During this offensive carry, an opposition player 'jumped' into the tackle with initial contact being made around the (injured) player's neck. The tackler's secondhand wrapped around the (injured) player's torso. The momentum for the 'jump tackle' resulted in the tackler's body being positioned behind the (injured) player, which halted the forward momentum of the (injured) player. As a result of the tackle, the (injured) player's right foot was fixed in external rotation. The (probable) injury event occurred when the opposition tackler 'collapsed' the (injured) player posteriorly to the ground resulting in forced dorsiflexion on a fixed externally rotated (right) foot. A similar mechanism of injury to the one described for case #9 was also observed in case #12 (indirect contact injury). In cases #4 and #8, the injury mechanism involved an externally induced valgus collapse of the knee joint coupled with pronation of the foot and a fixed externally rotated ankle; in both cases, the (injured) foot was 'pinned' in external rotation due to the weight of the tackling player falling onto the (injured) player's ankle. In case #4 (direct contact injury), the (injured) player was tackled by two opposition players; the tackle from the posterior-lateral direction resulted in a valgus collapse of the (injured) player's right knee joint. During this valgus collapse, his right foot was 'pinned' in pronation and external rotation as he was collapsed forward towards the ground during the tackle. A similar mechanism of injury to the one described for case #4 was also observed in case #8 (direct contact injury). In case #6 (direct contact injury), the (injured) player's (left) foot was 'locked' in external rotation by an 'ankle tackle' from an opposition player. At the same time, he was also tackled by another opposition team player. The (probable) injury event occurred as his ankle was forced into dorsiflexion while he was collapsed forward by the tacklers.

DISCUSSION

This is the first reported systematic video analysis to describe the mechanisms of acute ankle syndesmosis injuries in male professional rugby union players. The acute ankle syndesmosis injuries included in this study were sourced from the Leinster Rugby injury surveillance database and match/training video database. All included injuries were direct contact or indirect contact injuries, which were sustained while tackling or being tackled. Six of the injuries (46%) occurred as a result of an external load application to the injured limb typically via a tackle by an opposition player or due to inadvertent contact by a team-mate. In the other seven instances (54%), the injury occurred due to a collapse of the injured player's body mass over the injured ankle joint while tackling or being tackled.

Lower limb biomechanical profile at the index frame (the probable injury event)

The most common biomechanical presentation of the injured limb at the index frame (the probable injury event) was characterised by hip flexion, knee flexion, knee valgus, ankle dorsiflexion, pronation and external rotation of the foot. These results support previous assertions that the most common features of acute ankle syndesmosis injury include external rotation of the foot, eversion of the talus within the ankle mortise and excessive dorsiflexion.[14]

The ankle syndesmosis ligaments ensure the stability of the distal tibiofibular articulation by resisting the axial, rotational and translational forces that attempt to separate the tibia and fibula.[17] The anterior inferior tibiofibular ligament is a primary restraint to external rotation of the fibula, whereas the posterior inferior tibiofibular ligament is a primary restraint to posterior translation of the fibula.[18] The interosseous ligament, which merges with the interosseous membrane proximally, prevents lateral translation of the fibula.[18] During external rotation of the foot, the fibula rotates externally and translates posteriorly,[19] whereas the talus externally rotates producing a widening of the ankle joint mortise. Thus, forced external rotation of the foot is likely to result in a sprain of the anterior inferior and posterior inferior tibiofibular ligaments. Forceful eversion of the talus as occurs during weight-bearing pronation also produces a widening of the ankle joint mortise, with the potential to result in syndesmotic ligament sprain.[14] These ligaments are also likely to be sprained during forced dorsiflexion, during which the anterior aspect of the talar dome produces a widening of the distal tibiofibular articulation.[14]

Understanding the sport-specific mechanisms of injury associated with prevalent musculoskeletal injuries in a particular sport is integral to the planning of a thorough clinical assessment. Clinicians should be suspicious of acute ankle syndesmosis injury if a patient reports an injury mechanism including any of the following in isolation or in combination: external rotation of the foot; forced pronation; forced dorsiflexion of the ankle joint.

Injuries incurred while tackling: suboptimal tackle mechanics

The tackle is considered the most dangerous facet of rugby union game play, accounting for up to 58% of all injuries incurred during competitive matches.[20] In the present study, 6 of the 13 injuries (46%) were incurred while tackling an opposition player. In all instances, the tackle mechanics of the (injured) player were suboptimal.

In a descriptive epidemiological study on tackle injuries in professional rugby union, Quarrie and Hopkins[20] reported instances whereby suboptimal tackle technique was a key factor contributing to the occurrence of injury. Our data support the findings of Quarrie and Hopkins,[20] whereby suboptimal tackle technique was implicated in the mechanism of all acute syndesmosis injuries incurred while tackling. This is particularly evident in case #5 (indirect contact injury) and case #11 (indirect contact injury). In both cases, the (injured) player 'dropped' to a one-knee position while making a front-on tackle. It has been reported that a key determinant of successful outcomes in rugby union tackles is momentum (ie, product of a player's body mass and his velocity), with low momentum into a tackle being associated with a higher risk of injury.[21] In

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both case #5 (indirect contact injury) and case #11 (indirect contact injury), the tackler's (ie, injured player) momentum was effectively zero as he had no horizontal velocity as a result of dropping to a one-knee position. The characteristic features of suboptimal tackle technique associated with case #5 included: poor footwork resulting in a lack of control in his approach to the tackle; a wide lateral stance with neither leg pointed up-field—diminishing his ability to react efficiently to the movements of the ball carrier; falling into the tackle and dropping to a knee—resulting in a stationary absorption of the ball carrier's force through his kneeling limb. The characteristic feature of suboptimal tackle technique associated with case #7 included: the tackler over-running the ball carrier—resulting in his inability to remain square in the ruck, creating a vulnerable and exposed position of the (injured) limb to external forces; stationary absorption of the ball carrier's force through the injured limb. The characteristic features of suboptimal tackle technique associated with case #10 included: uncoordinated and incongruous actions by two connected tacklers; a poorly executed tackle by the first tackler—reaching, dropping to a knee with an inability to apply force to the ball carrier; lateral movement by the (injured) tackler—resulting in an inability to adjust to the height required to execute a successful tackle; stationary absorption of forces through the injured limb. The characteristic features of suboptimal tackle technique associated with case #11 included: a poor angle of approach—resulting in a decision making advantage for the ball carrier; a wide lateral stance with neither leg pointed up-field—resulting in an inability to react to the movements of the ball carrier; falling into the tackle and dropping to a knee—resulting in an inability to apply force and subsequent stationary absorption of the ball carrier's force. These cases highlight the importance of continued coaching of optimal tackle technique, even among professional rugby union players.

Injuries incurred while being tackled

Seven of the 13 injuries (54%) were incurred while being tackled by an opposition player. Four of the cases (cases #2, #3, #9, #12) were indirect contact injuries with the other three cases (cases #4, #6 and #8) being direct contact injuries. In three of these cases (cases #4, #9 and #12) the opposition player (tackler) 'jumped' into the tackle. Two of these cases (cases #9 and #12; both indirect contact injuries) resulted in the (injured) player being collapsed backwards over a fixed externally rotated foot. The injury mechanism in both cases also included forced dorsiflexion. In case #9 (direct contact injury), the tackler's body mass was loaded on the (injured) player's right ankle joint, which was 'pinned' in pronation and external rotation as he was collapsed forward towards the ground during the tackle. Each of these 'jump' tackle cases resulted in a high-grade sprain of the ball carrier's (injured player's) anterior inferior tibiofibular ligament with a concomitant sprain of the posterior inferior tibiofibular ligament. These data concur with the observations of Quarrie and Hopkins,[20] who reported that '*in instances where the tackler jumped on the ball carrier from the side or behind and the ball carrier attempted to continue running, the ball carrier appeared to be at particular risk of severe knee, lower leg, and ankle injuries*'. In the majority of cases, the tackler made contact with the (injured) player from a posterior direction. This observation concurs with the findings of previous research which noted that the ball carrier is at a heightened risk of injury if tackled from behind.[20]

What are the findings?

- The most common biomechanical presentation of the injured limb included a combination of hip flexion, knee flexion, knee valgus, ankle dorsiflexion, pronation and external rotation of the foot.
- Direct contact injuries typically occurred via the application of an external load to the injured limb—typically via a tackle by an opposition player or inadvertent contact by a team-mate.
- Indirect contact injuries typically involved the collapse of the injured player's body mass over the injured ankle joint while tackling or being tackled.
- Suboptimal tackle technique was often implicated in the injury mechanism.

How might it impact on clinical practice in the future?

- Clinicians should be suspicious of acute ankle syndesmosis injury if a patient reports an injury mechanism including any of the following in isolation or in combination: external rotation of the foot; forced pronation of the foot; forced dorsiflexion of the ankle joint.
- Continued coaching of optimal tackle technique is important, even among professional rugby union players.

Limitations

Our systematic video analysis of 13 acute ankle syndesmosis injuries were all from the same professional rugby union club. More than half of the injuries identified via the injury surveillance database could not be included in the analyses—primarily due to the index frame (probable injury event) not being visible. Future research with larger samples and fewer missing data are required to confirm or refute our findings. The inclusion of 13 cases is lower than other studies that have used systematic video analysis to detail the mechanism of musculoskeletal injuries in elite-level sports,[6 7 15 16 22] however, it must be noted that the injury incidence rate of acute ankle syndesmosis injuries is lower than lateral ankle sprains and adductor strains. The video quality, resolution and angle were limitations to this study. In some cases, the exact moment when a syndesmosis injury occurred could not be accurately determined, therefore the time point identified as the index frame (the probable injury event) was an estimate based on the subjective analysis of the three analysts. To overcome this issue, future studies should, if possible, engage players in the research process as they may be able to provide valuable input to help better detail the mechanisms of injury.

CONCLUSIONS

All of the acute ankle syndesmosis injuries included in the systematic video analysis were either direct or indirect contact injuries. The injuries incurred while tackling were the result of suboptimal tackle mechanics; therefore, continued efforts towards coaching optimal tackle mechanics are warranted. A majority of injuries incurred while being tackled involved a posterior tackle, which often resulted in a posterior collapse of the ball carrier's (injured player's) centre of mass over a fixed externally rotated ankle joint.

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Contributors ED, GF, AB, FK and CB co-developed the idea for the study. AB and PT sourced data for the study. FK developed the injury assessment form and ED and GF approved it. ED, AB, GF and HH analysed all the videos and reached consensus on the details of the injury assessment forms. FK mediated the consensus meeting. ED developed the qualitative description of each injury case (supplementary file 1). ED and AB drafted the initial manuscript, which as edited and agreed upon by all authors.

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REFERENCES

1 Fuller CW, Laborde F, Leather RJ, *et al*. International rugby board rugby world cup 2007 injury surveillance study. *Br J Sports Med* 2008;42:452–9.
2 Fuller CW, Sheerin K, Targett S. Rugby world cup 2011: international rugby board injury surveillance study. *Br J Sports Med* 2013;47:1184–91.
3 Fuller CW, Taylor A, Kemp SPT, *et al*. Rugby world cup 2015: world rugby injury surveillance study. *Br J Sports Med* 2017;51:51–7.
4 Sankey RA, Brooks JHM, Kemp SPT, *et al*. The epidemiology of ankle injuries in professional rugby union players. *Am J Sports Med* 2008;36:2415–24.
5 Bahr R, Krosshaug T. Understanding injury mechanisms: a key component of preventing injuries in sport. *Br J Sports Med* 2005;39:324–9.
6 Andersen TE, Floerenes TW, Arnason A, *et al*. Video analysis of the mechanisms for ankle injuries in football. *Am J Sports Med* 2004;32:69–79.
7 Skazalski C, Kruczynski J, Bahr MA, *et al*. Landing-related ankle injuries do not occur in plantarflexion as once thought: a systematic video analysis of ankle injuries in world-class volleyball. *Br J Sports Med* 2018;52:74–82.
8 Fong DT-P, Hong Y, Shima Y, *et al*. Biomechanics of supination ankle sprain: a case report of an accidental injury event in the laboratory. *Am J Sports Med* 2009;37:822–7.
9 Kristianslund E, Bahr R, Krosshaug T. Kinematics and kinetics of an accidental lateral ankle sprain. *J Biomech* 2011;44:2576–8.
10 Gehring D, Wissler S, Mornieux G, *et al*. How to sprain your ankle - a biomechanical case report of an inversion trauma. *J Biomech* 2013;46:175–8.
11 Terada M, Gribble PA. Jump landing biomechanics during a laboratory recorded recurrent ankle sprain. *Foot Ankle Int* 2015;36:842–8.
12 Mok K-M, Fong DT-P, Krosshaug T, *et al*. An ankle joint model-based image-matching motion analysis technique. *Gait Posture* 2011;34:71–5.
13 Fong DT-P, Ha SC-W, Mok K-M, *et al*. Kinematics analysis of ankle inversion ligamentous sprain injuries in sports: five cases from televised tennis Competitions. *Am J Sports Med* 2012;40:2627–32.
14 Lin C-F, Gross MT, Weinhold P. Ankle syndesmosis injuries: anatomy, biomechanics, mechanism of injury, and clinical guidelines for diagnosis and intervention. *J Orthop Sports Phys Ther* 2006;36:372–84.
15 Waldén M, Krosshaug T, Bjørneboe J, *et al*. Three distinct mechanisms predominate in non-contact anterior cruciate ligament injuries in male professional football players: a systematic video analysis of 39 cases. *Br J Sports Med* 2015;49:1452–60.
16 Montgomery C, Blackburn J, Withers D, *et al*. Mechanisms of ACL injury in professional rugby Union: a systematic video analysis of 36 cases. *Br J Sports Med* 2018;52:994–1001.
17 Golanó P, Vega J, de Leeuw PAJ, *et al*. Anatomy of the ankle ligaments: a pictorial essay. *Knee Surg Sports Traumatol Arthrosc* 2010;18:557–69.
18 McCollum GA, van den Bekerom MPJ, Kerkhoffs GMMJ, *et al*. Syndesmosis and deltoid ligament injuries in the athlete. *Knee Surg Sports Traumatol Arthrosc* 2013;21:1328–37.
19 Beumer A, van Hemert WLW, Niesing R, *et al*. Radiographic measurement of the distal tibiofibular syndesmosis has limited use. *Clin Orthop Relat Res* 2004;423:227–34.
20 Quarrie KL, Hopkins WG. Tackle injuries in professional rugby Union. *Am J Sports Med* 2008;36:1705–16.
21 Garraway WM, Lee AJ, Macleod DA, *et al*. Factors influencing tackle injuries in rugby Union football. *Br J Sports Med* 1999;33:37–41.
22 Serner A, Mosler AB, Tol JL, *et al*. Mechanisms of acute adductor longus injuries in male football players: a systematic visual video analysis. *Br J Sports Med* 2019;53:158–64.

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